

LUCIDMSK CME PROGRAM

MODULE 03 — POST-TEST

Shoulder MRI: Rotator Cuff, Instability & Arthroplasty

Shoulder · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)[™]

Read each question carefully and select the single best answer. A score of $\geq 70\%$ (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1 Parasagittal T1 MRI shows fat content equal to muscle in supraspinatus. Goutallier grade:

- A Grade 1
- B Grade 2
- C Grade 3
- D Grade 4

2 Supraspinatus stump retracted to the glenoid level. Patte stage:

- A Stage 1
- B Stage 2
- C Stage 3
- D Cannot determine from MRI

3 Critical Shoulder Angle (CSA) measures 38°. Increased risk for:

- A Glenohumeral OA
- B Supraspinatus tear
- C Biceps rupture
- D AC joint arthritis

4 Axial MRI shows posterior humeral head subluxation with biconcave glenoid and >2.5 mm step-off. Walch type:

- A A2
- B B1
- C B2
- D B3

5

rTSA CT shows inferior scapular neck erosion contacting the inferior baseplate screw. Sirveaux notching grade:

- A Grade 1
- B Grade 2
- C Grade 3
- D Grade 4

6

Hamada Grade 3 has which ADDITIONAL finding compared to Grade 2?

- A AHI ≤ 5 mm
- B GH joint narrowing
- C Acetabularization of the acromion
- D Humeral head collapse

7

Post-aTSA "rocking horse" sign at the glenoid-cement interface indicates:

- A Periprosthetic fracture
- B Glenoid component loosening
- C Rotator cuff tear
- D Periprosthetic infection

8

Which Goutallier grade is a relative contraindication to rotator cuff repair, especially infraspinatus?

- A Grade 0
- B Grade 1
- C Grade 2
- D Grade 3–4

9

Seebauer Type IIB rotator cuff arthropathy is characterized by:

A	Centered, stable; intact CA arch
B	Centered, medialized; early bone loss
C	Decentered; limited escape; acromion erosion
D	Anterior escape; AC joint eroded; most destabilized

10 A supraspinatus stump remnant of 18 mm is identified after retraction. Surgical significance:

A	No significance
B	Stump >15 mm predicts repairability (Meyer 2012)
C	Stump <20 mm indicates irreversible atrophy
D	Stump length only relevant for subscapularis

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	C	C	B	C	B	C	B	D	D	B

Q1 Parasagittal T1 MRI shows fat content equal to muscle in supraspinatus. Goutallier grade: **Answer: C**

A Grade 1

B Grade 2

✓ **Grade 3**

D Grade 4

EXPLANATION Grade 3 = fat equal to muscle. Grade 2 = fat less than muscle. Grade 4 = fat exceeds muscle. Grade ≥ 2 predicts poorer rotator cuff repair outcomes.

Q2 Supraspinatus stump retracted to the glenoid level. Patte stage: **Answer: C**

A Stage 1

B Stage 2

✓ **Stage 3**

D Cannot determine from MRI

EXPLANATION Patte Stage 3 = stump at or beyond glenoid level. Stage 2 = stump at humeral head level. Stage 1 = near greater tuberosity. Stage 3 tears may not be primarily repairable.

Q3 Critical Shoulder Angle (CSA) measures 38° . Increased risk for: **Answer: B**

A Glenohumeral OA

✓ **Supraspinatus tear**

C Biceps rupture

D AC joint arthritis

EXPLANATION CSA $>35^\circ$ = increased supraspinatus tear risk — high acromion creates superior shear on the cuff. CSA $<30^\circ$ predicts glenohumeral OA. Normal range = $30\text{--}35^\circ$.

Q4 Axial MRI shows posterior humeral head subluxation with biconcave glenoid and >2.5 mm step-off. Walch type: **Answer: C**

A A2

B B1

✓ **B2**

D B3

EXPLANATION Walch B2 = posterior subluxation + biconcavity + >2.5 mm step-off. B1 lacks biconcavity. B3 = monoconcave posterior wear with >15° retroversion. A2 = centered with central erosion.

Q5

rTSA CT shows inferior scapular neck erosion contacting the inferior baseplate screw. Sirveaux notching grade:

Answer: B

A Grade 1

✓ **Grade 2**

C Grade 3

D Grade 4

EXPLANATION Sirveaux Grade 2 = notch contacts the inferior screw. Grade 1 stays within the pillar. Grade 3 exceeds the screw affecting baseplate stability. Grade 4 extends to the inferior baseplate border.

Q6

Hamada Grade 3 has which ADDITIONAL finding compared to Grade 2?

Answer: C

A AHI ≤5 mm

B GH joint narrowing

✓ **Acetabularization of the acromion**

D Humeral head collapse

EXPLANATION Grade 3 adds acetabularization (concavity/remodeling of the CA arch) to Grade 2's superior migration (AHI ≤5 mm). Grade 4 adds GH narrowing. Grade 5 = head collapse.

Q7

Post-aTSA "rocking horse" sign at the glenoid-cement interface indicates:

Answer: B

A Periprosthetic fracture

✓ **Glenoid component loosening**

C Rotator cuff tear

D Periprosthetic infection

EXPLANATION "Rocking horse" sign = radiolucent lines at the glenoid-cement-bone interface with component tilting — classic sign of glenoid loosening due to failed bone ingrowth or cement mantle failure.

Q8

Which Goutallier grade is a relative contraindication to rotator cuff repair, especially infraspinatus?

Answer: D

A Grade 0

B Grade 1

C Grade 2

✓ **Grade 3–4**

EXPLANATION Goutallier ≥3 (fat ≥ muscle) is a strong contraindication, particularly for infraspinatus. Grade ≥2 infraspinatus atrophy predicts repair failure — rTSA should be considered.

Q9**Seebauer Type IIB rotator cuff arthropathy is characterized by:****Answer: D**

- A Centered, stable; intact CA arch
- B Centered, medialized; early bone loss
- C Decentered; limited escape; acromion erosion

✓ **Anterior escape; AC joint eroded; most destabilized**

EXPLANATION Seebauer IIB = anterior escape with extensive CA arch, AC joint, and clavicle destruction — most destabilized pattern, highest rTSA complication risk. Type IA = centered and stable (most favorable).

Q10**A supraspinatus stump remnant of 18 mm is identified after retraction. Surgical significance:****Answer: B**

- A No significance
- ✓ **Stump >15 mm predicts repairability (Meyer 2012)**
- C Stump <20 mm indicates irreversible atrophy
- D Stump length only relevant for subscapularis

EXPLANATION Meyer DC et al. (2012): remnant stump >15 mm predicts successful repair and tissue hold. Report stump length with Patte stage and Goutallier grade together for complete surgical assessment.